# PARTIAL CIRCUIT MILESTONE — NOT A RELEASE DESIGN

Global nets join sheets. GEN_ and HUB_ segregate control and phase nets. DC_LINK / PGND shared.

36–58.4 V screening range only. No validated current rating. No PCB or routing.

Protected entry still missing; OV chopper and fault memory populated on dedicated pages.

GEN DRV8353S driver and three low-side current channels

File: gen-driver.kicad_sch

GEN 100 V MOSFET bridge — schematic only

File: gen-bridge.kicad_sch

HUB DRV8353S driver and three low-side current channels

File: hub-driver.kicad_sch

HUB 100 V MOSFET bridge — schematic only

File: hub-bridge.kicad_sch

Unfinished safety / control / supply boundaries

File: external-boundaries.kicad_sch

Bus-fed auxiliary and independent safety supply

File: aux-supply.kicad_sch

Autonomous OV chopper comparator and rail / brake monitors

File: protection-comparators.kicad_sch

Release-before-rearm DFF memory

File: fault-latch.kicad_sch

LVC Schmitt fault inputs without series diode offsets

File: fault-interfaces.kicad_sch

Independent brake chopper power stage

File: brake-chopper.kicad_sch

STM32G474VET6 LQFP100: power, reset, debug and control

File: mcu-core.kicad_sch

MCU 3V3 decoupling — external supervised rail

File: mcu-power.kicad_sch

Six CSA ADC input filters and brake status sense

File: mcu-analog.kicad_sch

3V3 Hall interfaces and optional quadrature encoder

File: mcu-sensors.kicad_sch

GEN independent raw CSA window trips

File: gen-fast-oc.kicad_sch

HUB independent raw CSA window trips

File: hub-fast-oc.kicad_sch

Ratiometric OC window and wired sink fault

File: fast-oc-reference.kicad_sch

Bounded bench DC entry — manual dead-circuit sequencing only

File: dc-entry.kicad_sch

LEFT supervised binary brake interface

File: brake-intent-left.kicad_sch

RIGHT supervised binary brake interface

File: brake-intent-right.kicad_sch

Rider dump PWM OR independent emergency overvoltage

File: brake-dump-command.kicad_sch

OFFBOARD supervised switch end-of-line network — candidate

File: brake-switch-harness.kicad_sch

LEFT selected FSR low-voltage bench frontend

File: fsr-left.kicad_sch

LEFT FSR candidate ESD and secondary clamps — unqualified

File: fsr-protection-left.kicad_sch

RIGHT selected FSR low-voltage bench frontend

File: fsr-right.kicad_sch

RIGHT FSR candidate ESD and secondary clamps — unqualified

File: fsr-protection-right.kicad_sch

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheets: /
File: powerboard-reva.kicad_sch

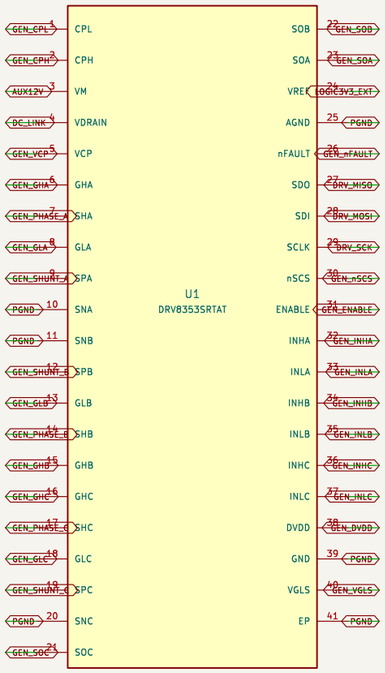
**Title: Pedal-by-wire: hierarchical engineering review**

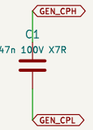
| Size: A1 | Date: | **Rev: A-PARTIAL** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 1/27 |

TI SLVSDY6A pp6–7 pin table. 40-pin RTA + exposed pad. SPI variant, NOT hardware variant.

VM supplied by bus-fed auxiliary 12 V. VREF=3V3 (CSA midpoint internally VREF/2).

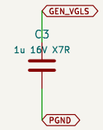
RELEASE BLOCKERS: firmware, OC timing/coverage and transient/thermal qualification unfinished.





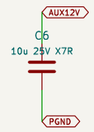








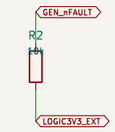


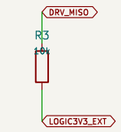



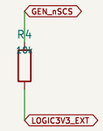







ENABLE pull-down defaults off; not a safety-rated inhibit. SPI configuration and CSA gain20 not yet validated.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

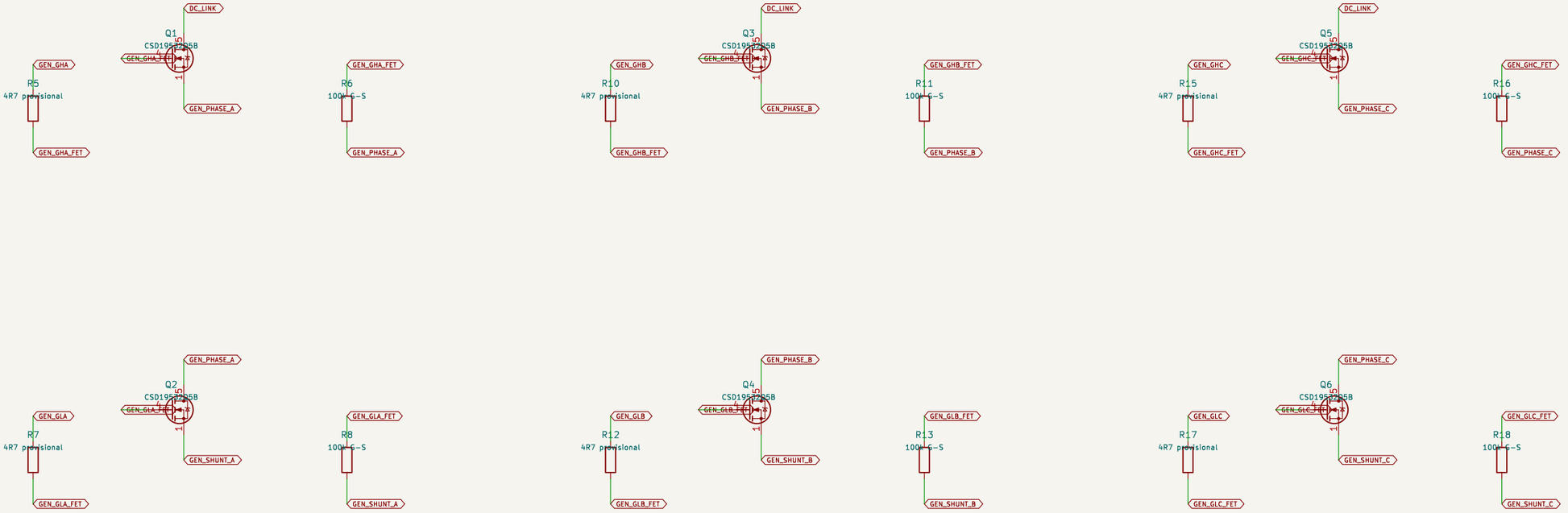
Sheet: /GEN DRV8353S driver and three low-side current channels/
File: gen-driver.kicad_sch

**Title: GEN DRV8353S driver and three low-side current channels**

| Size: A3 | Date: | **Rev: A-PARTIAL** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 2/27 |

CSD19532Q5B stock KiCad symbol / pad map. Gates: 4; drains: 5; sources: 1,2,3.

Labels are electrical connections. Rgate=4R7 provisional. Kelvin shunts: no routing qualified.




Shunt amplifier SPx taps SHUNT_x; SNx taps PGND: Kelvin layout mandatory. 15A RMS => 0.45W per shunt.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /GEN 100 V MOSFET bridge — schematic only/
File: gen-bridge.kicad_sch

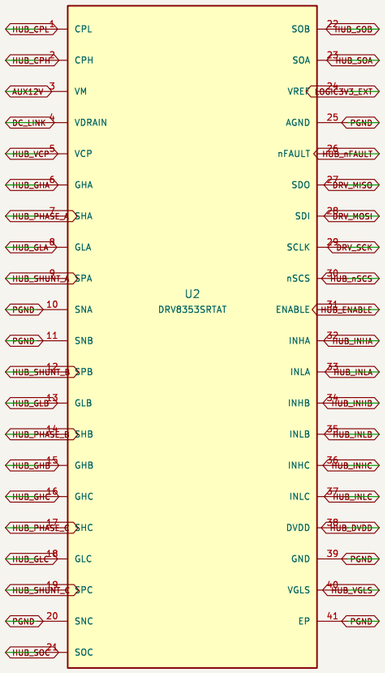
**Title: GEN 100 V MOSFET bridge — schematic only**

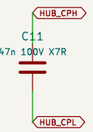
| Size: A3 | Date: | **Rev: A-PARTIAL** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 3/27 |

TI SLVSDY6A pp6–7 pin table. 40-pin RTA + exposed pad. SPI variant, NOT hardware variant.

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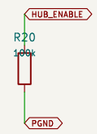


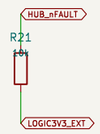


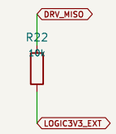


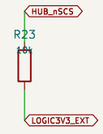





ENABLE pull-down defaults off; not a safety-rated inhibit. SPI configuration and CSA gain20 not yet validated.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

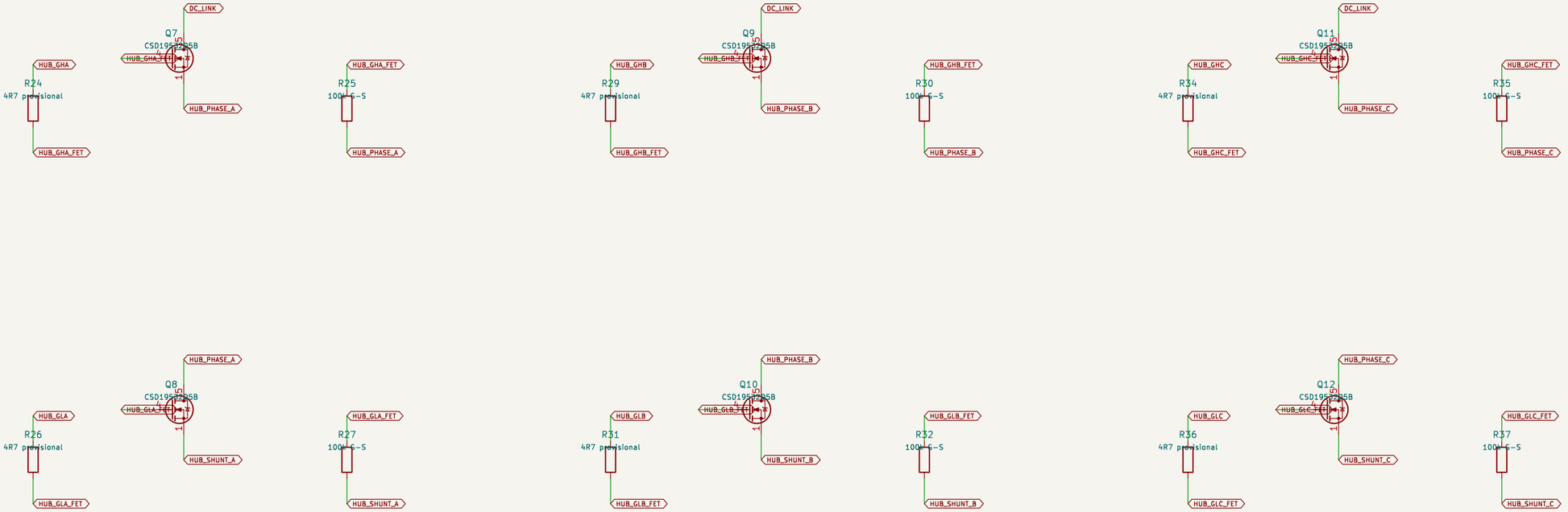
Sheet: /HUB DRV8353S driver and three low-side current channels/
File: hub-driver.kicad_sch

**Title: HUB DRV8353S driver and three low-side current channels**

| Size: A3 | Date: | **Rev: A-PARTIAL** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 4/27 |

CSD19532Q5B stock KiCad symbol / pad map. Gates: 4; drains: 5; sources: 1,2,3.

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Sheet: /HUB 100 V MOSFET bridge — schematic only/
File: hub-bridge.kicad_sch

**Title: HUB 100 V MOSFET bridge — schematic only**

| Size: A3 | Date: | **Rev: A-PARTIAL** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 5/27 |

OFFBOARD interfaces: 3V3 supply, brake loop and protected bus entry.

DO NOT ENERGIZE. Protected entry unfinished; MCU and protection remain unqualified.

J1
BUS_TEST_ONLY_NOT_SOURCE_INPUT
DC_LINK
PGND

J2
MCU_SUPPLY_EXT
LOGIC3V3_EXT
PGND

J3
GEN_PHASE_A
GEN_PHASE_B
GEN_PHASE_C

J4
HUB_PHASE_A
HUB_PHASE_B
HUB_PHASE_C

J5
SAFETY_FAULT_CHAIN_NC
SAFETY3V3
FAULT_CHAIN

DC_LINK
C21
100u 100V provisional
PGND

Missing: source fuse, reverse blocking, precharge, DC contactor, bus current sensor and discharge.

Populated: OV chopper, fault latch, brake loop, watchdog, aux supply. Added: raw CSA window trip; timing/coverage unqualified.

MCU and sensor interfaces populated; firmware, thermal sensors, storage/BMS acceptance missing.

PGND is common, not galvanically isolated. Supply sequencing and back-power paths unqualified.

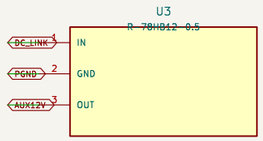
NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

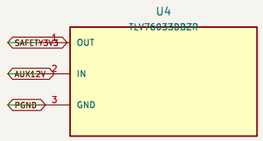
Sheet: /Unfinished safety / control / supply boundaries/
File: external-boundaries.kicad_sch

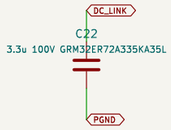
**Title: Unfinished safety / control / supply boundaries**

| Size: A3 | Date: | Rev: A-PARTIAL |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 6/27 |

R-78HB12-0.5: 17–72V operating, 75V absolute. Entry transient protection still REQUIRED.



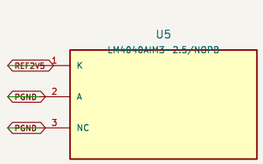






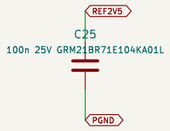













Safety supply is NOT the MCU supply. LOGIC3V3_EXT remains a supervised, external MCU rail.

Supply total/current/thermal budget provisional; no galvanic isolation or stored-energy ride-through.

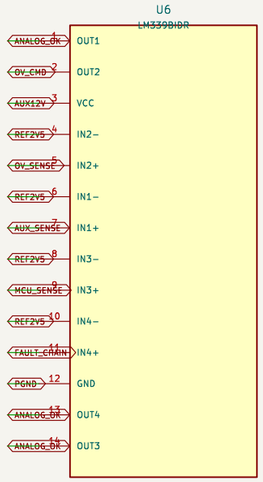
NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

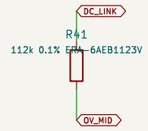
Sheet: /Bus-fed auxiliary and independent safety supply/
File: aux-supply.kicad_sch

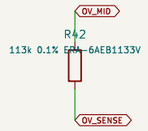
**Title: Bus-fed auxiliary and independent safety supply**

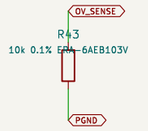
| Size: A3 | Date: | Rev: A-PARTIAL |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 7/27 |

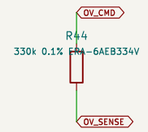
Comparator powered from AUX12V; open collectors pulled to SAFETY3V3. No MCU in OV path.



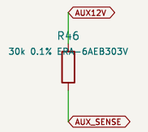



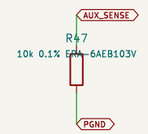



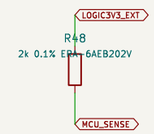



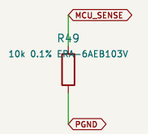


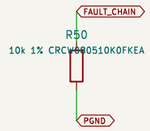




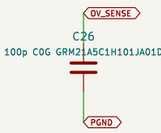













FAULT_CHAIN must close to SAFETY3V3 through normally-closed E-stop + dump thermal switches (NOT rider levers). Open wire trips.

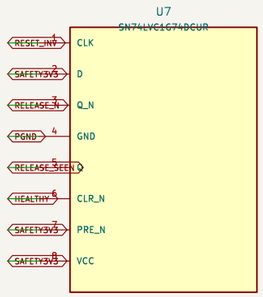
NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

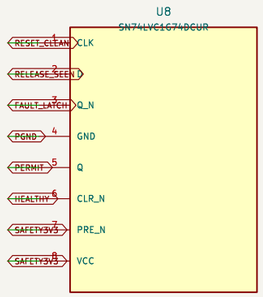
Sheet: /Autonomous OV chopper comparator and rail / brake monitors/
File: protection-comparators.kicad_sch

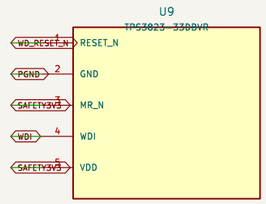
**Title: Autonomous OV chopper comparator and rail / brake monitors**

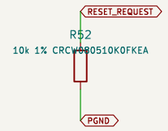
| | | |
|---|---|---|
| Size: A3 | Date: | **Rev: A-PARTIAL** |
| KiCad E.D.A. 9.0.9 | | Id: 8/27 |

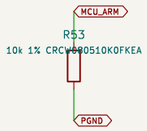
UNSAFE DRAFT: healthy release edge THEN press edge required. No automatic reset.











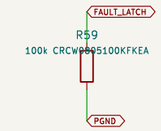


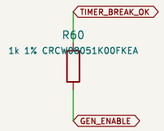


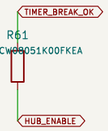






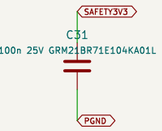









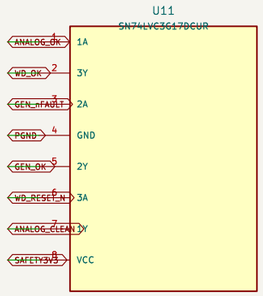

TPS3823 non-A: falling WDI only; hold low during RESET. Stuck RESET stays disabled.

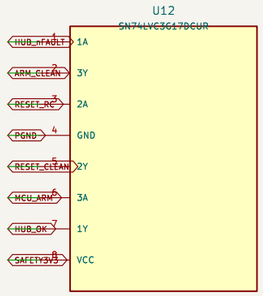
NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

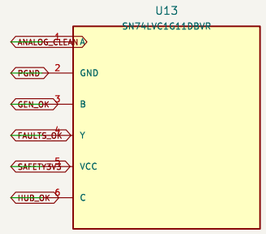
Sheet: /Release-before-rearm DFF memory/
File: fault-latch.kicad_sch

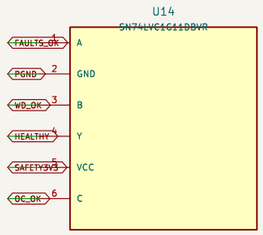
**Title: Release-before-rearm DFF memory**

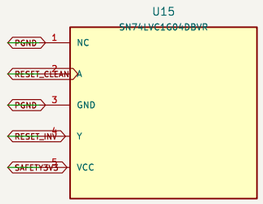
| Size: A3 | Date: | **Rev: A-PARTIAL** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 9/27 |

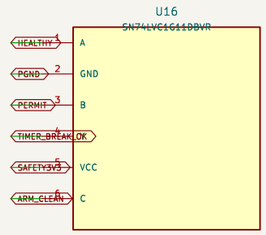





















Default OFF via async supervisor clear. Analog ramp/metastability NOT validated.

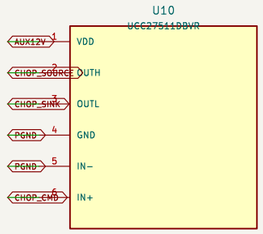
NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

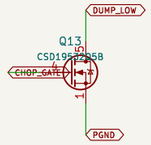
Sheet: /LVC Schmitt fault inputs without series diode offsets/
File: fault-interfaces.kicad_sch

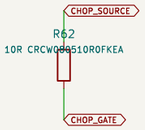
**Title: LVC Schmitt fault inputs without series diode offsets**

| Size: A3 | Date: | Rev: A-PARTIAL |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 10/27 |

Autonomous OV path ignores bridge ENABLE. External resistor is required across BRAKE_DUMP_RESISTOR connector pins.

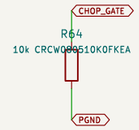


















Candidate: 2 x TE HSC3004R7J in series (9.4 ohm). Chassis/heatsink + NC thermal cutoff required.

No resistor cable clamp / fuse / SOA qualification yet. Loss of AUX supply loses bus sink: NOT FAIL SAFE.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

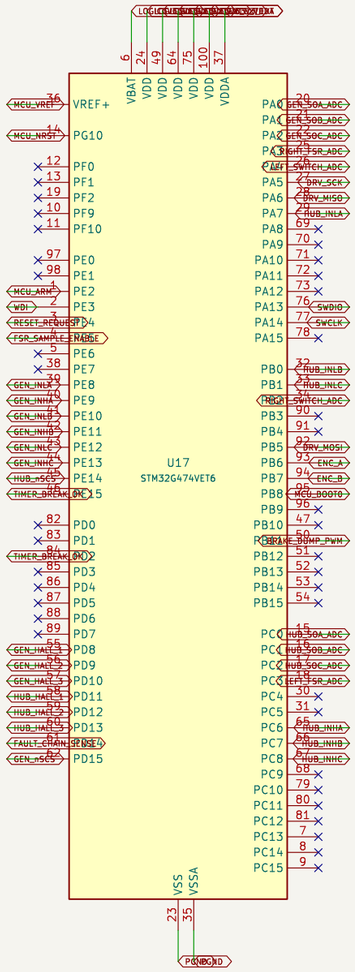
Sheet: /Independent brake chopper power stage/
File: brake-chopper.kicad_sch

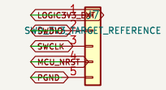
**Title: Independent brake chopper power stage**

| Size: A3 | Date: | **Rev: A-PARTIAL** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 11/27 |

HSI16 internal clock / PLL option: no HSE or LSE fitted; PF0/PF1 and PC14/15 explicit NC.

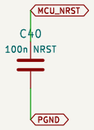
BOOT0 low; keep NRST function and configure boot option bytes before use. No firmware supplied.

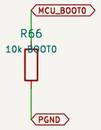












BKIN active LOW required in TIM1/TIM8. Separate hardware ENABLE path remains intact.

NC GPIO: leave analog mode; HSI accuracy / PLL / ADC timing require validation.

MCU_ARM / RESET_REQUEST 10k and WDI 1k default-low resistors are on fault-latch page.

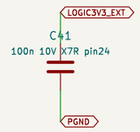
NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

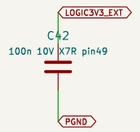
Sheet: /STM32G474VET6 LQFP100: power, reset, debug and control/
File: mcu-core.kicad_sch

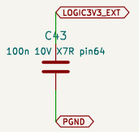
**Title: STM32G474VET6 LQFP100: power, reset, debug and control**

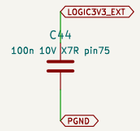
| Size: A3 | Date: | **Rev: A-PARTIAL** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 12/27 |

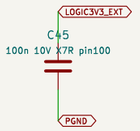
One 100n per VDD pin (24,49,64,75,100) plus VBAT pin6. Place each at pin / ground return.

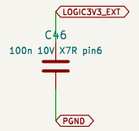


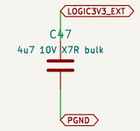



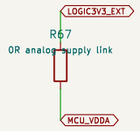



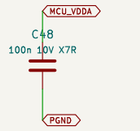



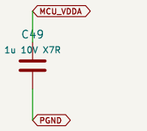



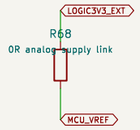



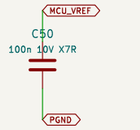



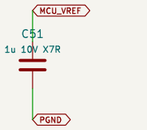




VREFBUF MUST remain disabled: external 3V3 reference. VDDA/VREF tied via OR; not precision isolated.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

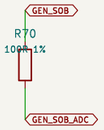
Sheet: /MCU 3V3 decoupling — external supervised rail/
File: mcu-power.kicad_sch

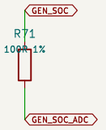
**Title: MCU 3V3 decoupling — external supervised rail**

| | | |
|---|---|---|
| Size: A3 | Date: | **Rev: A-PARTIAL** |
| KiCad E.D.A. 9.0.9 | | Id: 13/27 |

100R / 1nF candidate filters: settling / sample aperture / CSA gain remain unqualified.



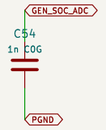





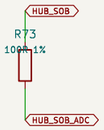


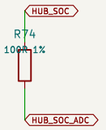




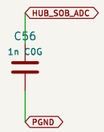


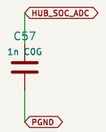


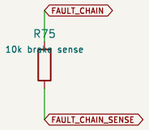



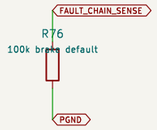


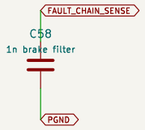





FAULT_CHAIN_SENSE is status ONLY. E-stop / thermal comparator and fault path do not depend on firmware.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /Six CSA ADC input filters and brake status sense/
File: mcu-analog.kicad_sch

**Title: Six CSA ADC input filters and brake status sense**

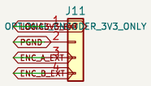
| Size: A3 | Date: | **Rev: A-PARTIAL** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 14/27 |

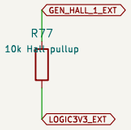
3V3 ONLY: open-collector Hall / encoder or compatible 3V3 outputs. NO 5V tolerance claim.







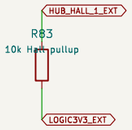






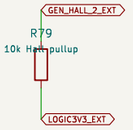














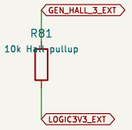


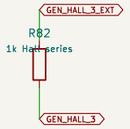


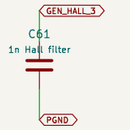








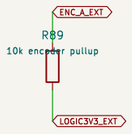






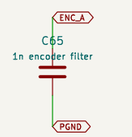


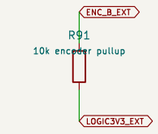


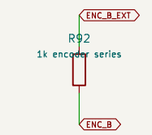


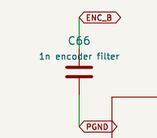







Encoder PB6/PB7: TIM4_CH1/CH2 AF2 DS12288r5 T13 p74. Cable ESD/EMC protection unresolved.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

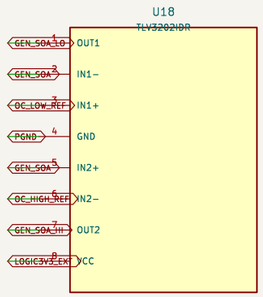
Sheet: /3V3 Hall interfaces and optional quadrature encoder/
File: mcu-sensors.kicad_sch

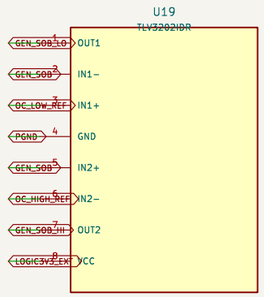
**Title: 3V3 Hall interfaces and optional quadrature encoder**

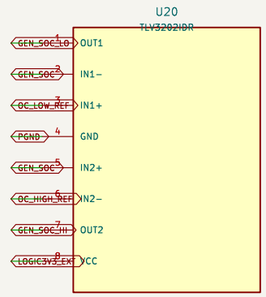
| Size: A3 | Date: | **Rev: A-PARTIAL** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 15/27 |

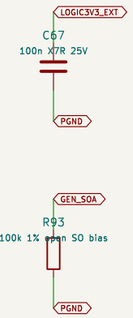
TLV3202IDR: positive SP−SN current DECREASES SO. Lower and upper limits both latch off.

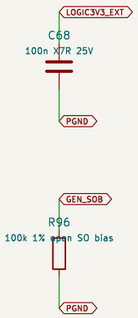
No PWM blanking added. Low−side blind intervals remain; DRV VDS OCP contract mandatory.



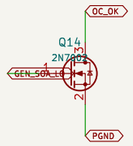



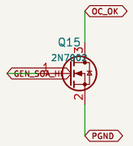



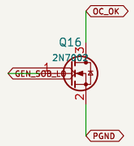



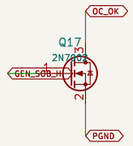


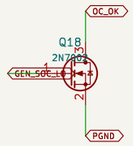


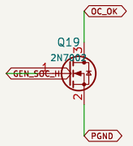


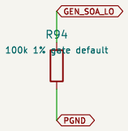



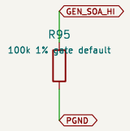



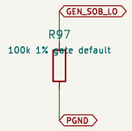



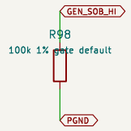



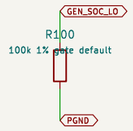



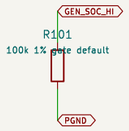




Push−pull comparators drive separate sink FETs; outputs are NOT shorted together.

Open SO pulls down; stuck midpoint / shunt open / power−loss safety NOT proven. UNSAFE DRAFT.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

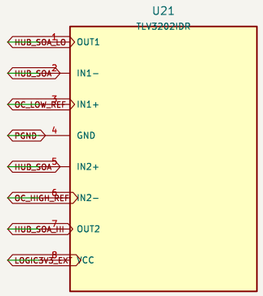
Sheet: /GEN independent raw CSA window trips/
File: gen−fast−oc.kicad_sch

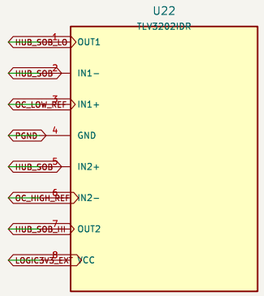
**Title: GEN independent raw CSA window trips**

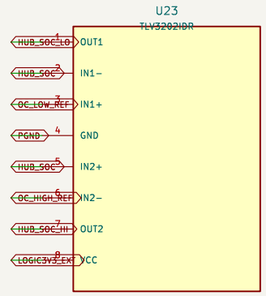
| Size: A3 | Date: | **Rev: A−PARTIAL** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 16/27 |

TLV3202IDR: positive SP−SN current DECREASES SO. Lower and upper limits both latch off.

No PWM blanking added. Low−side blind intervals remain; DRV VDS OCP contract mandatory.

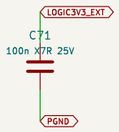


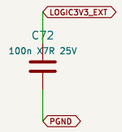



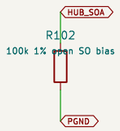



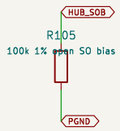


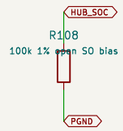


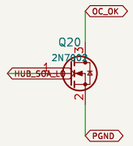


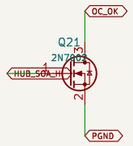



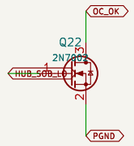



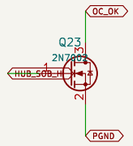



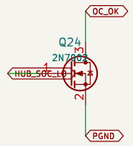



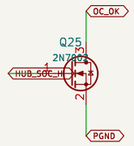



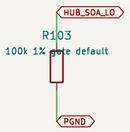



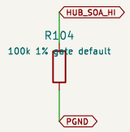



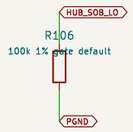



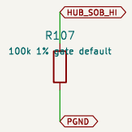



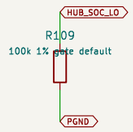



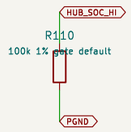




Push−pull comparators drive separate sink FETs; outputs are NOT shorted together.

Open SO pulls down; stuck midpoint / shunt open / power−loss safety NOT proven. UNSAFE DRAFT.

NOT SAFE TO ENERGIZE − SAFETY CIRCUITS UNFINISHED

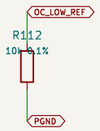
Sheet: /HUB independent raw CSA window trips/
File: hub−fast−oc.kicad_sch

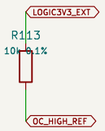
**Title: HUB independent raw CSA window trips**

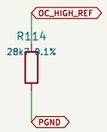
| Size: A3 | Date: | **Rev: A−PARTIAL** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 17/27 |

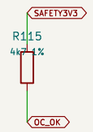
VREF tracks actual driver LOGIC3V3_EXT: 0.852713 / 2.447287 V nominal at 3.3 V.




OC_OK directly enters U14.6, clears U7/U8 and U16: both ENABLE and TIM1/TIM8 break.

No external hysteresis or blanking: first crossing latches. Noise/settling and sink delay unqualified.

Nominal trip +/−19.932 A; provisional measurement +/−30 A. Commissioning cap 10 A, NOT release.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

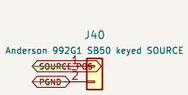
Sheet: /Ratiometric OC window and wired sink fault/
File: fast-oc-reference.kicad_sch

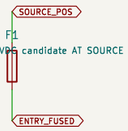
**Title: Ratiometric OC window and wired sink fault**

| Size: A3 | Date: | **Rev: A-PARTIAL** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 18/27 |

NOT SAFE TO ENERGIZE integrated drive. No automatic sequencing or reverse-blocking guarantee.

36-58.4V bench interface, 15A TARGET ONLY. Disconnect PM machines before source removal.

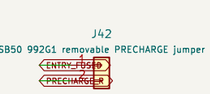


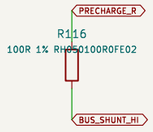



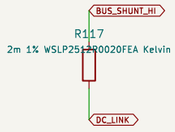


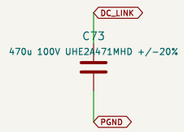


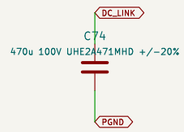


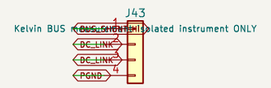










NO approved jumper sequence. Never manipulate J40/J41/J42 energized or bus charged.

J43 raw high common mode: NOT MCU ADC. D20/fuse reverse-fault coordination UNQUALIFIED.

Sink/chopper stays DC_LINK side. MAIN open does NOT make bus safe: PM backfeeds via diodes.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

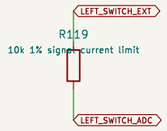
Sheet: /Bounded bench DC entry – manual dead-circuit sequencing only/
File: dc-entry.kicad_sch

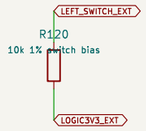
**Title: Bounded bench DC entry – manual dead-circuit sequencing only**

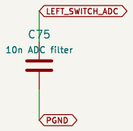
| Size: A3 | Date: | **Rev: A-PARTIAL** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 19/27 |

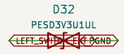
Candidate only: reversible non-load-bearing lever sensors. NO mechanical brake cable alteration.

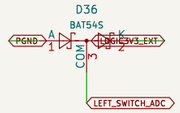



External switch harness: 4k7 series + (10k parallel NC switch). Release=1.055V; press=1.964V.

Open=3V3 / short=0V faults via ADC; firmware NOT implemented. Clamp injection/power-off NOT qualified.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

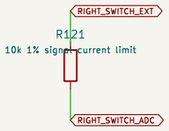
Sheet: /LEFT supervised binary brake interface/
File: brake-intent-left.kicad_sch

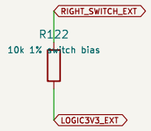
**Title: LEFT supervised binary brake interface**

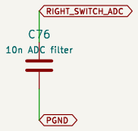
| Size: A3 | Date: | **Rev: A-PARTIAL** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 20/27 |

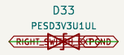
Candidate only: reversible non-load-bearing lever sensors. NO mechanical brake cable alteration.

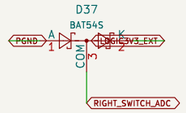



External switch harness: 4k7 series + (10k parallel NC switch). Release=1.055V; press=1.964V.

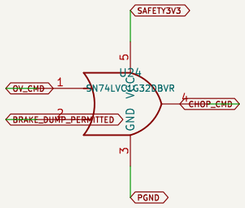
Open=3V3 / short=0V faults via ADC; firmware NOT implemented. Clamp injection/power-off NOT qualified.

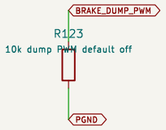
NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /RIGHT supervised binary brake interface/
File: brake-intent-right.kicad_sch

**Title: RIGHT supervised binary brake interface**

| Size: A3 | Date: | **Rev: A-PARTIAL** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 21/27 |





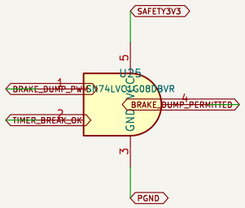


OV always overrides MCU PWM. Normal PWM must require valid rider request AND cool dump permission.

PWM AND TIMER_BREAK_OK: watchdog / E-stop / thermal fault kill normal dump. OV bypass retained.

No torque-sign hardware guard. Firmware must disarm on sensor fault; watchdog / OC / E-stop remain hardware kills.

Firmware entirely unimplemented. Do not energize; brake command control only a design contract.



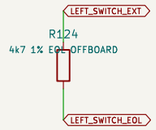


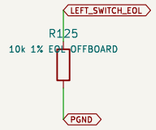


NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

| | | |
|---|---|---|
| Sheet: /Rider dump PWM OR independent emergency overvoltage/<br>File: brake-dump-command.kicad_sch | | |
| **Title: Rider dump PWM OR independent emergency overvoltage** | | |
| Size: A3 | Date: | **Rev: A-PARTIAL** |
| KiCad E.D.A. 9.0.9 | | Id: 22/27 |

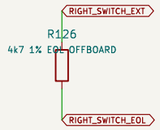
OFFBOARD parts must live at lever, not board. NC contact opens on rider braking. No brake cable modification.

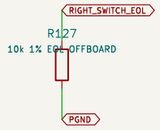












Switch symbol drawn open (braking). Model cables unplugged by removing BOTH EOL resistors and switch.

Welded contact or resistor short can imitate release; FSR plausibility is firmware only, not independent safety.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

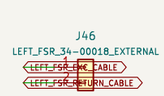
Sheet: /OFFBOARD supervised switch end-of-line network — candidate/
File: brake-switch-harness.kicad_sch

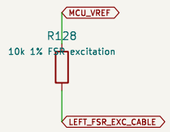
**Title: OFFBOARD supervised switch end-of-line network — candidate**

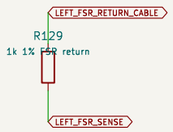
| Size: A3 | Date: | **Rev: A-PARTIAL** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 23/27 |

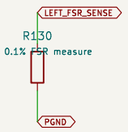
Interlink 34-00018 external 402 short-tail pad; connector mate / mechanics UNQUALIFIED.

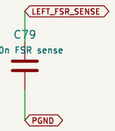
LOW-VOLTAGE BENCH ONLY. TVS / clamps candidates, NOT battery-short protection. No torque-sign guard.

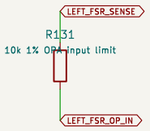


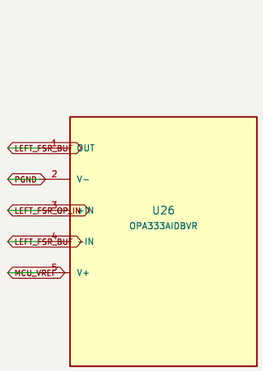



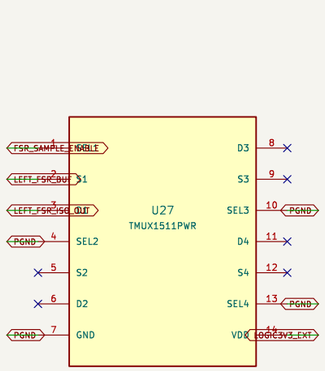



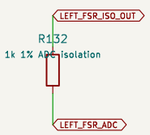



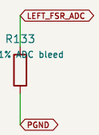


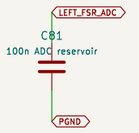


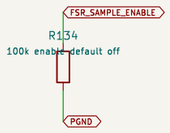


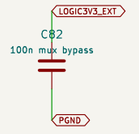




MCU_VREF = excitation / OPA supply; LOGIC3V3_EXT = mux supply. PGND common; no isolation of ground.

Enable PE5 pin4 default low; 20ms blanking + plausibility pending firmware / bench validation.

Binary switch authoritative by firmware CONTRACT ONLY. Invalid FSR cannot authorize positive torque.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

Sheet: /LEFT selected FSR low-voltage bench frontend/
File: fsr-left.kicad_sch

**Title: LEFT selected FSR low-voltage bench frontend**

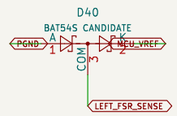
| Size: A3 | Date: | **Rev: A-PARTIAL** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 24/27 |

Connector TVS: cathode1 cable, anode2 ground. BAT54S: anode1 ground, cathode2 rail, junction3 sense.










NOT QUALIFIED: clamp voltage, leakage, negative pulses, rail sink / powered-off backfeed, ESD energy.

Battery short is a RELEASE BLOCKER. No external cable driver; segregated current-limited LV bench only.

TVS is NOT a 3.3V limiter. Secondary clamp can backfeed MCU_VREF. No vehicle energization authorization.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

Sheet: /LEFT FSR candidate ESD and secondary clamps – unqualified/
File: fsr-protection-left.kicad_sch

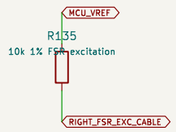
**Title: LEFT FSR candidate ESD and secondary clamps – unqualified**

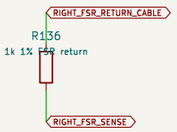
| Size: A3 | Date: | **Rev: A-PARTIAL** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 25/27 |

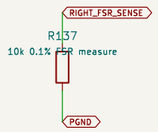
Interlink 34-00018 external 402 short-tail pad; connector mate / mechanics UNQUALIFIED.

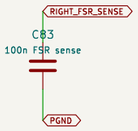
LOW-VOLTAGE BENCH ONLY. TVS / clamps candidates, NOT battery-short protection. No torque-sign guard.

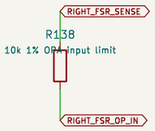


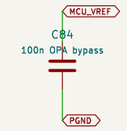



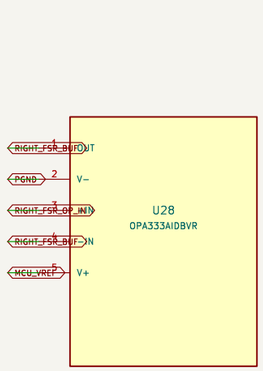



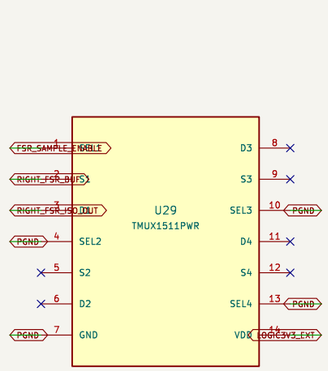



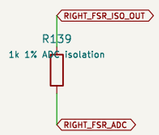



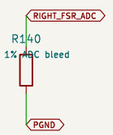



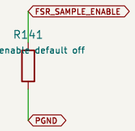



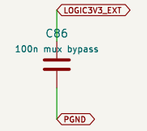










MCU_VREF = excitation / OPA supply; LOGIC3V3_EXT = mux supply. PGND common; no isolation of ground.

Enable PE5 pin4 default low; 20ms blanking + plausibility pending firmware / bench validation.

Binary switch authoritative by firmware CONTRACT ONLY. Invalid FSR cannot authorize positive torque.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

Sheet: /RIGHT selected FSR low-voltage bench frontend/
File: fsr-right.kicad_sch

**Title: RIGHT selected FSR low-voltage bench frontend**

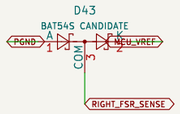
| Size: A3 | Date: | **Rev: A-PARTIAL** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 26/27 |

Connector TVS: cathode1 cable, anode2 ground. BAT54S: anode1 ground, cathode2 rail, junction3 sense.










NOT QUALIFIED: clamp voltage, leakage, negative pulses, rail sink / powered-off backfeed, ESD energy.

Battery short is a RELEASE BLOCKER. No external cable driver; segregated current-limited LV bench only.

TVS is NOT a 3.3V limiter. Secondary clamp can backfeed MCU_VREF. No vehicle energization authorization.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

Sheet: /RIGHT FSR candidate ESD and secondary clamps – unqualified/
File: fsr-protection-right.kicad_sch

**Title: RIGHT FSR candidate ESD and secondary clamps – unqualified**

| Size: A3 | Date: | **Rev: A-PARTIAL** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 27/27 |